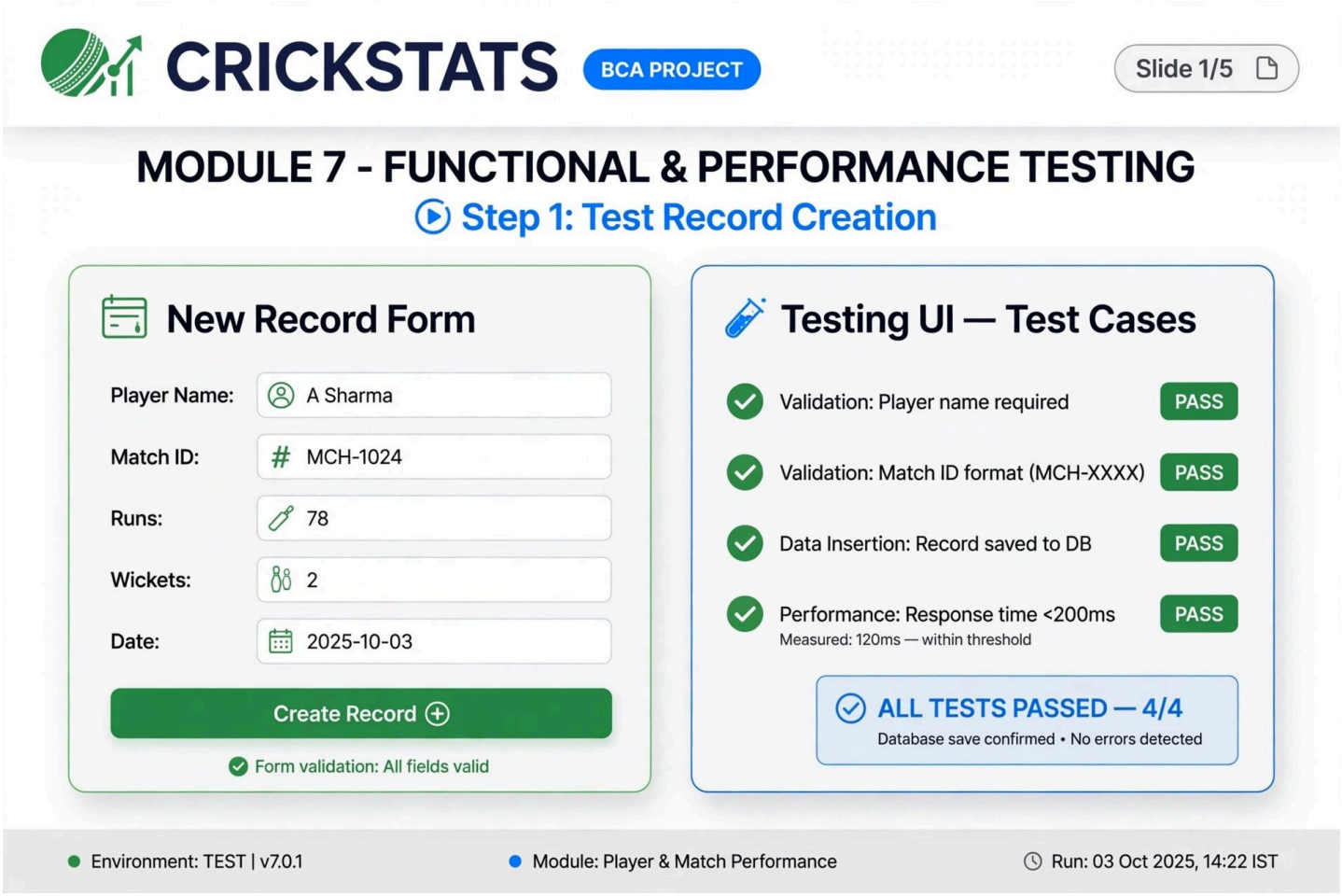

CRICKSTATS
BCA PROJECT
Slide 1/5
MODULE 7 - FUNCTIONAL & PERFORMANCE TESTING
Step 1: Test Record Creation
New Record Form
Player Name:
A Sharma
Match ID:
MCH-1024
Runs:
78
Wickets:
2
Date:
2025-10-03
Create Record
Form validation: All fields valid
Testing UI — Test Cases
Validation: Player name required
PASS
Validation: Match ID format (MCH-XXXX)
PASS
Data Insertion: Record saved to DB
PASS
Performance: Response time <200ms
Measured: 120ms — within threshold
PASS
ALL TESTS PASSED — 4/4
Database save confirmed • No errors detected
Environment: TEST | v7.0.1
Module: Player & Match Performance
Run: 03 Oct 2025, 14:22 IST

MODULE 7 - FUNCTIONAL & PERFORMANCE TESTING

Step 2: Mandatory Fields and Validation

Mandatory Fields *

Form Validation

* Player Name *	Virat Kohli
* Email Address *	player@example.com
* Match Date *	2024-03-15
* Runs Scored *	85

• All fields marked with * are mandatory and must be completed before submission

Validation Rules

Email Format Validation
Must be valid email format (name@domain.com)

Date Validation
Date cannot be in the future • Format: YYYY-MM-DD

Numeric Range Validation
Runs Scored: 0–999 • Must be whole number

Performance Check: Validation response time < 200ms
Passed
All validations execute instantly on field input

MODULE 7 - FUNCTIONAL & PERFORMANCE TESTING

Step 3: Automatic Updating of Statistics

Auto-Update Flow

Match Data Submitted
New match result entered ✓

→

Validation & Processing
Data validated •
• Calculations triggered

→

Stats Calculation Engine
Runs • Avg • Strike Rate • Total

→

Player Stats Updated
Database updated •
Live dashboard refresh

ⓘ Trigger: Automatic update occurs within 1–2 seconds after match submission
No manual refresh required

Before Update

Player: Virat Kohli • ID: CS-014

Batting Average:	45.20 –
Total Runs:	1,240 –
Strike Rate:	128.50 –

Last Updated: 14 Mar 2024 — 16:32

After Update

Player: Virat Kohli • ID: CS-014

Batting Average:	47.10 ↑ +1.90
Total Runs:	1,325 ↑ +85
Strike Rate:	131.20 ↑ +2.70

Last Updated: 14 Mar 2024 — 16:32:18 • Just now

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MODULE 7 - FUNCTIONAL & PERFORMANCE TESTING

Step 4: Roll-up Summary Calculations

Roll-up Summary Overview

Purpose
Verify aggregated statistics are calculated correctly across multiple matches for each player

Calculation Logic

Total Runs (Sum)
Sum = Match1 + Match2 + Match3 + ... + MatchN
42 + 85 + 63 = 190 runs

Batting Average
Average = Total Runs ÷ Total Dismissals
190 ÷ 4 = 47.50 average

Strike Rate
Strike Rate = (Total Runs ÷ Balls Faced) × 100
(190 ÷ 145) × 100 = 131.03 SR

All roll-up calculations recomputed automatically on new match entry

Verification Results

Player: Rohit Sharma • ID: CS-021

	Match	Runs	Status
1.	Match #12 vs MI	42	Verified
2.	Match #13 vs CSK	85	Verified
3.	Match #14 vs RCB	63	Verified

Roll-up Summary Calculated:

- **Total Runs:** 190 — Sum of all matches matches total
- **Average:** 47.50 — Calculation accurate (190/4)
- **Strike Rate:** 131.03 — Calculation accurate (190/145 ×100)

Verification Status: ALL CALCULATIONS PASSED
No discrepancies detected • Accuracy: 100%

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MODULE 7 - FUNCTIONAL & PERFORMANCE TESTING

Step 5: Report Generation and Dashboard Performance

Dashboard Performance Metrics

Avg Load Time
1.24s
✓ 42% faster than target (2s)

Uptime
99.97%
✓ Exceeds 99.9% SLA target

Concurrent Users
150+
✓ Tested successfully with no errors

Load Time Performance



ⓘ Target: <2.0s for all reports • All metrics pass performance criteria

Report Generation

PDF Export: Generated successfully
Avg generation time: 1.6s • File size: 240KB

CSV Export: Generated successfully
Avg generation time: 0.8s • 1,200 records exported

Scheduled Reports: Active
Daily at 23:00 • Automated email delivery to team

Report Accuracy: 100%
All data validated against database • No discrepancies found

Performance Test Result: PASSED
All functional & performance tests completed successfully